

Problem Set 2023

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Question 2

2.1 Economic interpretation of θ

θ is a crucial parameter in the intertemporal utility function that modifies the disutility of labor or the valuation of leisure. A high θ indicates a significant disutility associated with labor, reflecting either a strong preference for leisure or a high cost of labor in terms of utility. Conversely, a low θ suggests less valuation of leisure or a preference for labor.

2.2 Household objective optimisation problem

$$\max_{\{C_t, N_t, K_t\}} V_t = \frac{1}{1-\sigma} \cdot \left(C_t - \theta_t \cdot \frac{N_t^{1+\phi}}{1+\phi} \right)^{1-\sigma} + \beta \cdot \mathbb{E}_t [V_{t+1}] + \lambda_t \cdot [W_t \cdot N_t + R_{k,t} \cdot K_{t-1} - C_t - K_t + (1-\delta) \cdot K_{t-1}]$$

$$0 < \beta < 1, \quad 0 < \delta < 1, \quad \phi \geq 0, \quad \sigma > 1$$

- First Condition Order

$$C_t : \quad \lambda_t = \left(C_t - \theta_t \cdot \frac{N_t^{1+\phi}}{1+\phi} \right)^{-\sigma} \quad (1)$$

$$N_t : \quad \lambda_t \cdot W_t = \left(C_t - \theta_t \cdot \frac{N_t^{1+\phi}}{1+\phi} \right)^{-\sigma} \cdot \left(\theta_t \cdot N_t^\phi \right) \quad (2)$$

$$K_t : \quad \lambda_t = \beta \cdot \mathbb{E}_t [\lambda_{t+1} \cdot (1 + [R_{k,t+1} - \delta])] \quad (3)$$

- Euler's intertemporal equation (a): (1) $\rightarrow$ (3) and intratemporal Labor supply (b): (1) $\rightarrow$ (2)

$$1 = \beta \cdot \mathbb{E}_t \left[\left(\frac{C_{t+1} - \theta_{t+1} \cdot \frac{N_{t+1}^{1+\phi}}{1+\phi}}{C_t - \theta_t \cdot \frac{N_t^{1+\phi}}{1+\phi}} \right)^{-\sigma} \cdot (1 + [R_{k,t+1} - \delta]) \right] \quad (a)$$

$$W_t = \theta_t \cdot N_t^\phi \quad (b)$$

2.3 Firm's profit maximisation problem

$$\max_{\{K_{t-1}, N_t\}} \Pi = A_t \cdot K_{t-1}^\alpha \cdot N_t^{1-\alpha} - R_{k,t} \cdot K_{t-1} - W_t \cdot N_t$$

- First Condition Order

$$K_{t-1} : \quad R_{k,t} = \alpha \cdot A_t \cdot \left(\frac{K_{t-1}}{N_t} \right)^{\alpha-1} \quad (i)$$

$$N_t : \quad W_t = (1-\alpha) \cdot A_t \cdot \left(\frac{K_{t-1}}{N_t} \right)^\alpha \quad (ii)$$

2.4 Steady Stead

$$A = 1, \quad \theta = 1$$

- Euler's Equation (a) evaluated at the steady state

$$1 = \beta \cdot \left(\frac{C - \theta \cdot \frac{N^{1+\phi}}{1+\phi}}{C - \theta \cdot \frac{N^{1+\phi}}{1+\phi}} \right)^{-\sigma} \cdot [1 + (R_k - \delta)]$$

$$R_k = \frac{1}{\beta} - 1 + \delta \quad \begin{cases} \beta = 0,96 \\ \delta = 0,075 \end{cases}$$

$$R_k = \frac{1}{0,96} - 1 + 0,075$$

$$R_k = 0,116667$$

- α , (i) and R_k to find the value of $\frac{K}{N}$

$$R_k = \alpha \cdot A \cdot \left(\frac{K}{N} \right)^{\alpha-1}$$

$$\frac{K}{N} = \left(\frac{\alpha \cdot A}{R_k} \right)^{\frac{1}{1-\alpha}} \quad \begin{cases} A = 1 \\ \alpha = 0,33 \\ R_k = 0,116667 \end{cases}$$

$$\frac{K}{N} = \left(\frac{0,33 \cdot 1}{0,116667} \right)^{\frac{1}{1-0,33}}$$

$$\mathcal{K}_N = 4,720422$$

- $\mathcal{K}_N$ and (ii) to find the value of W

$$W = (1 - \alpha) \cdot A \cdot (\mathcal{K}_N)^\alpha \quad \begin{cases} \alpha = 0,33 \\ \mathcal{K}_N = 4,720422 \end{cases}$$

$$W = (1 - 0,33) \cdot 1 \cdot (4,720422)^{0,33}$$

$$W = 1,11812$$

- W , θ , ϕ and (b) to find the value of N

$$W = \theta \cdot N^\phi$$

$$N = \left(\frac{W}{\theta} \right)^{\frac{1}{\phi}} \quad \begin{cases} \phi = 4 \\ \theta = 1 \\ W = 1,11812 \end{cases}$$

$$N = \left(\frac{1,11812}{1} \right)^{\frac{1}{4}}$$

$$N = 1,028305$$

- $\mathcal{K}_N$ and N to find the value of K

$$\mathcal{K}_N = \frac{K}{N}$$

$$K = \mathcal{K}_N \cdot N \quad \begin{cases} \mathcal{K}_N = 4,720422 \\ N = 1,028305 \end{cases}$$

$$K = (4,720422) \cdot (1,028305)$$

$$K = 4,854034$$

- K, N, α to find the value of Y

$$Y = K^\alpha \cdot N^{1-\alpha} \quad \left\{ \begin{array}{l} \alpha = 0,33 \\ K = 4,854034 \\ N = 1,028305 \end{array} \right.$$

$$Y = (4,854034)^{0,33} \cdot (1,028305)^{1-0,33}$$

$$Y = 1,716072$$

- δ, K, N, R_K, W to find the value of C

$$C + K - K \cdot (1 - \delta) = W \cdot N + R_K \cdot K$$

$$C = W \cdot N + (R_K - \delta) \cdot K \quad \left\{ \begin{array}{l} \delta = 0,075 \\ K = 4,854034 \\ N = 1,028305 \\ R_K = 0,116667 \\ W = 1,11812 \end{array} \right.$$

$$C = (1,11812) \cdot (1,028305) + (0,116667 - 0,075) \cdot (4,854034)$$

$$C = 1,352021$$

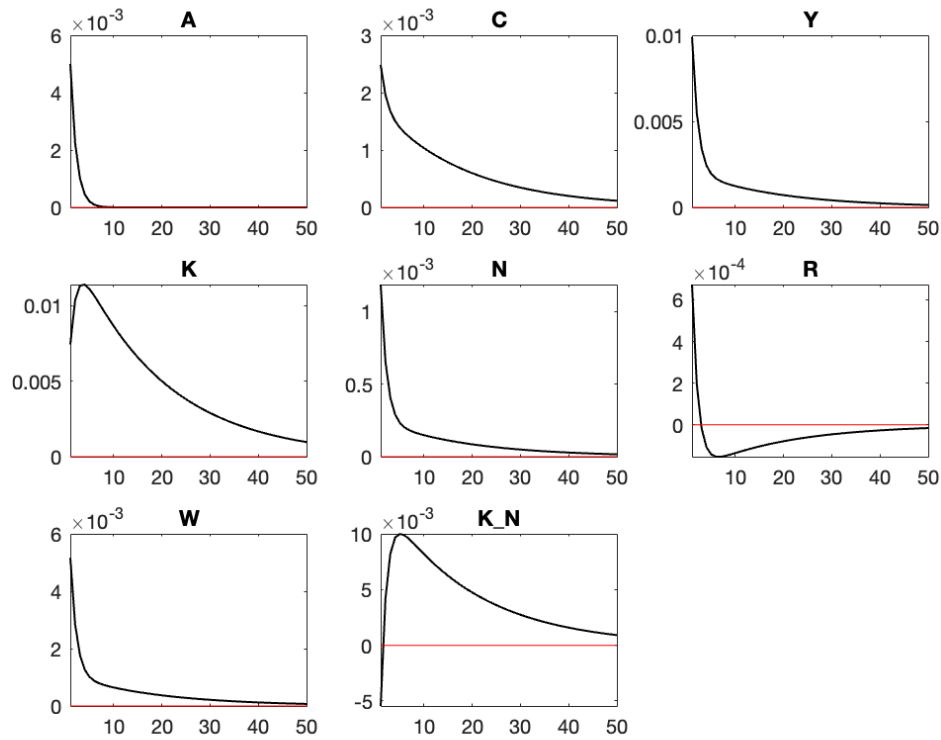


Figure 1

2.5 Economic explanation of the IRF's shape on Figure 1

- A_t : A positive technology shock, indicated by an increase in ε_t^A , results in a direct and immediate uplift in the technology level A_t , enhancing productivity. This boost is temporary, as A_t returns to its steady state, following a pattern of gradual decline.
- Y_t : The surge in A_t leads to an increase in economic output Y_t . This is due to the enhanced efficiency and productivity in the production process of the firms brought by the improved technology level.
- W_t and $R_{k,t}$: The shock positively influences the marginal productivity of labor and capital. Consequently, wages W_t increase due to higher labor productivity, and the return on capital $R_{k,t}$ rises. This leads to an increase in the real interest rate R_t , which is the return on capital adjusted for depreciation ($R_{k,t} - \delta$), reflecting improved capital efficiency.
- C_t and N_t : Higher wages lead to increased household income, which in turn boosts consumption C_t . Simultaneously, the attractiveness of higher wages leads to an increase in labor supply N_t .
- K_t : Triggered by the higher return on capital $R_{k,t}$, there's an incentive for increased savings, promoting capital accumulation K_t . This growth in capital stock continues until the influence of the technology shock diminishes.

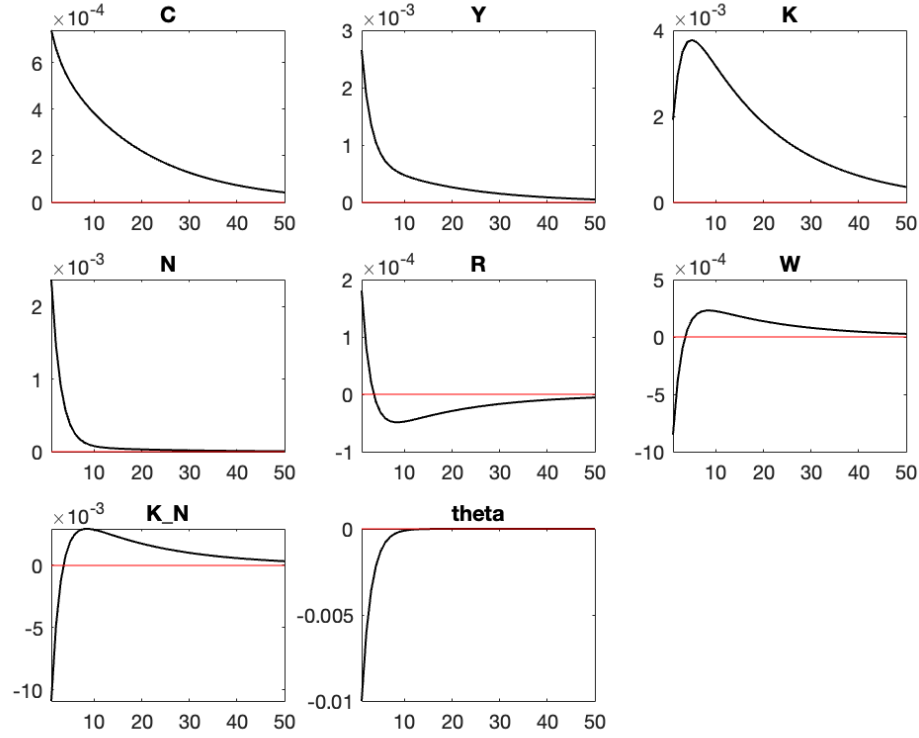


Figure 2

2.6 Economic explanation of the IRF's shape following a negative preference shock Figure 2

- θ_t : The parameter θ_t represents the household's relative preference for leisure over consumption. A negative shock in ε_t^θ leads to a decrease in θ_t , symbolizing a sudden shift towards greater preference for consumption relative to leisure. This change prompts the household to allocate more time to labor, given the reduced disutility of work. The decline in θ_t is temporary, and it gradually converges back to its steady state, reflecting the household's reversion to its original preference balance over time.
- N_t : The lower disutility of labor, as a consequence of the shock, leads to an increase in labor supply N_t . This change reflects the household's willingness to work more for a given wage level, driven by the altered preference structure.
- W_t : In response to the rise in N_t and the resulting fall in marginal labor productivity, firms adjust wages downward. The decrease in wages W_t aligns with the new equilibrium in the labor market, as the cost of labor adapts to its increased supply.
- R_t : The shock induces an increase in the productivity of capital, leading to a rise in the real interest rate R_t . This upswing in R_t signals a more profitable environment for capital investment, prompting an accumulation of capital K_t .
- K_t : The enhanced return on capital encourages savings and investment, resulting in a buildup of capital K_t . This accumulation persists as long as the real interest rate remains positive, reinforcing the economy's productive capacity.
- Y_t : The simultaneous increase in both capital K_t and labor N_t drives an upturn in output Y_t . This boost in production is a natural outcome of the augmented input factors, reflecting a more robust economic output.
- C_t : The rise in output and the initial increase in labor supply contribute to a higher level of consumption C_t . The household's ability to consume more is supported by the increased income from the additional labor supplied, despite the downward wage adjustment.

2.7 Comparison of the Real Wage (W_t) IRF Direction in *Figure 1* and *Figure 2*

- The shocks exert distinct initial influences on the economy. Specifically, the real wage W_t reacts differently to technology and preference shocks. A positive technology shock leads to increased marginal productivity of labor, prompting firms to raise wages to match the improved labor efficiency. In contrast, a negative preference shock causes a rise in labor supply due to households favoring work over leisure, leading to an excess of labor. This excess supply on the labor market diminishes the bargaining power of workers, compelling firms to lower wages due to the decreased marginal productivity of labor.